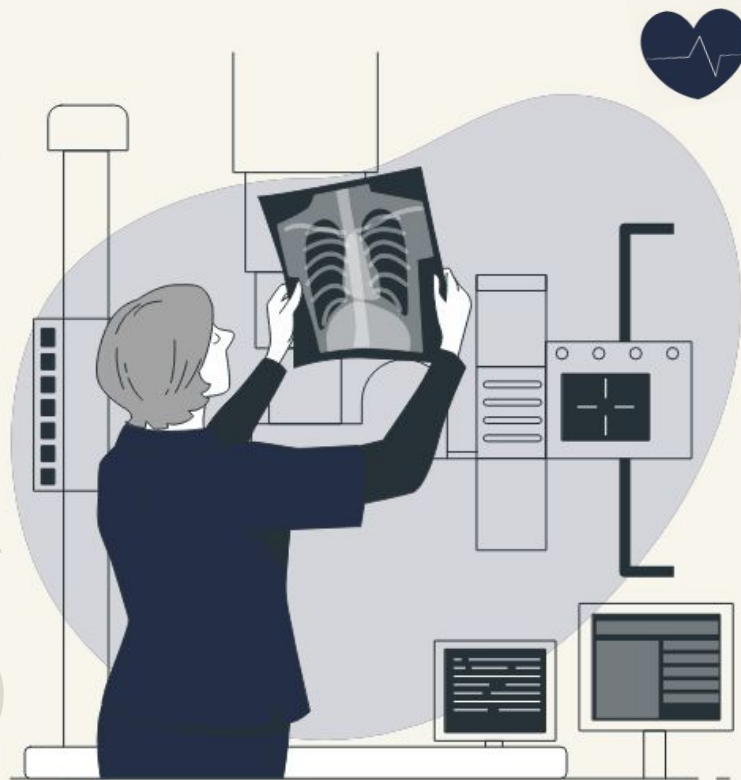


npower Chest-Detect

A Cloud-Native AI Diagnostic Pipeline for
Automated Chest Imaging



Group 2

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01

Background



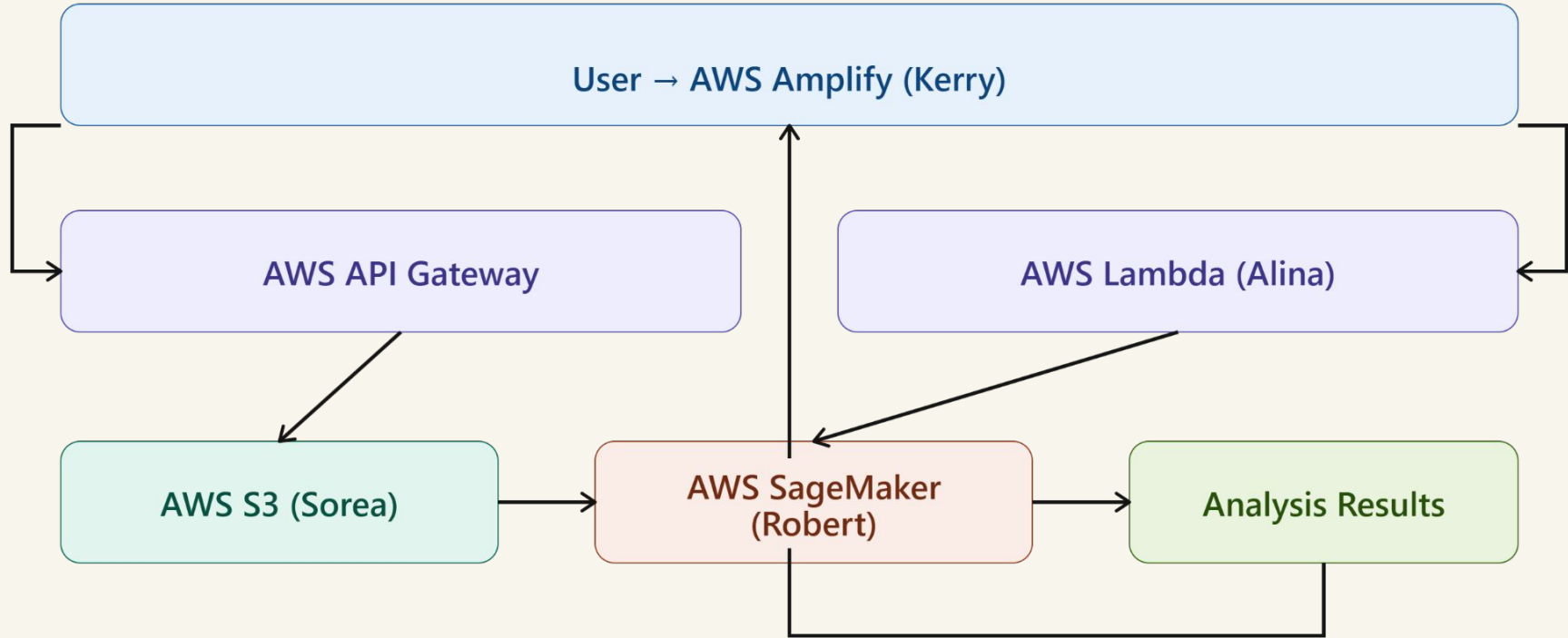




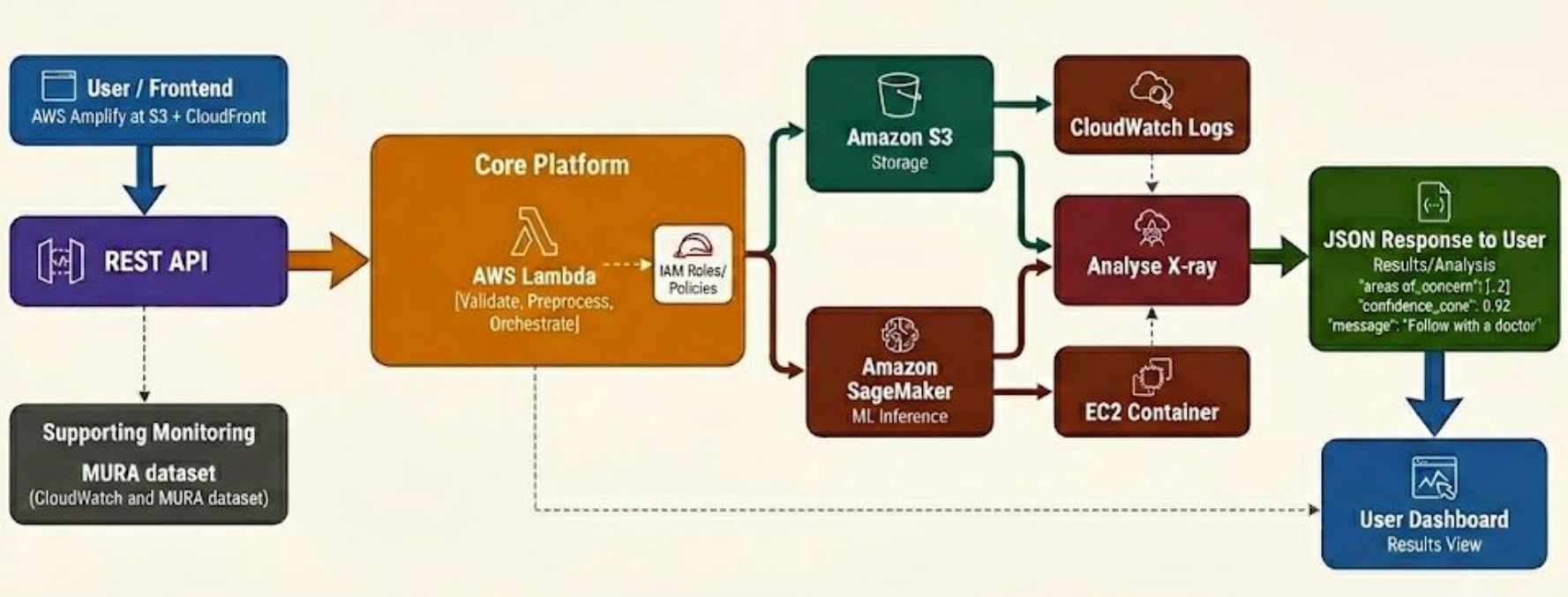
Mission Statement

High-volume **chest X-ray** pipelines are heavily bottlenecked & lack automated, point-of-capture AI tools, **leaving patients** waiting in **uncertainty over** complex **results** they cannot easily understand. To bridge this gap, **our machine learning** inference API provides rapid, preliminary anomaly **detection** to empower patients & **prompt faster**, proactive **follow-ups** with their doctors.

AWS CloudWatch (Noe: Logs & Monitoring)

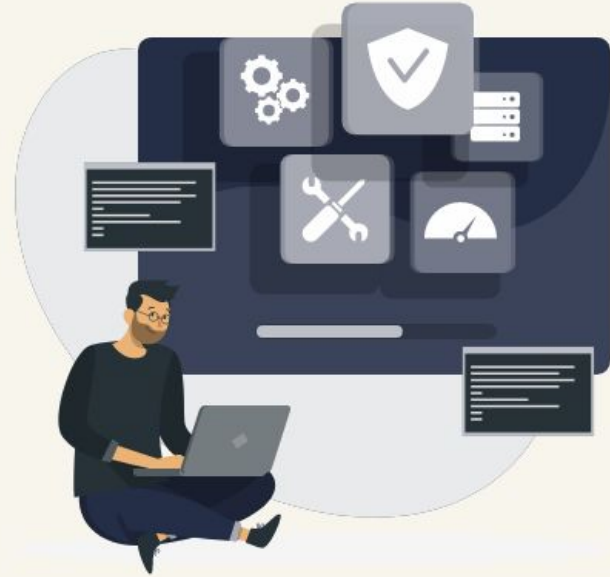


Architecture Overview



02

Technical Breakdown



Project Roles

01

Data & Storage Engineer

Sorea Asmai

02

Frontend Engineer

Kerry Bell

03

Machine Learning Engineer

Sorea Asmai

04

Backend Engineer

Alina Varentsova

05

QA Tester & Orchestrator

Noe Fonseca



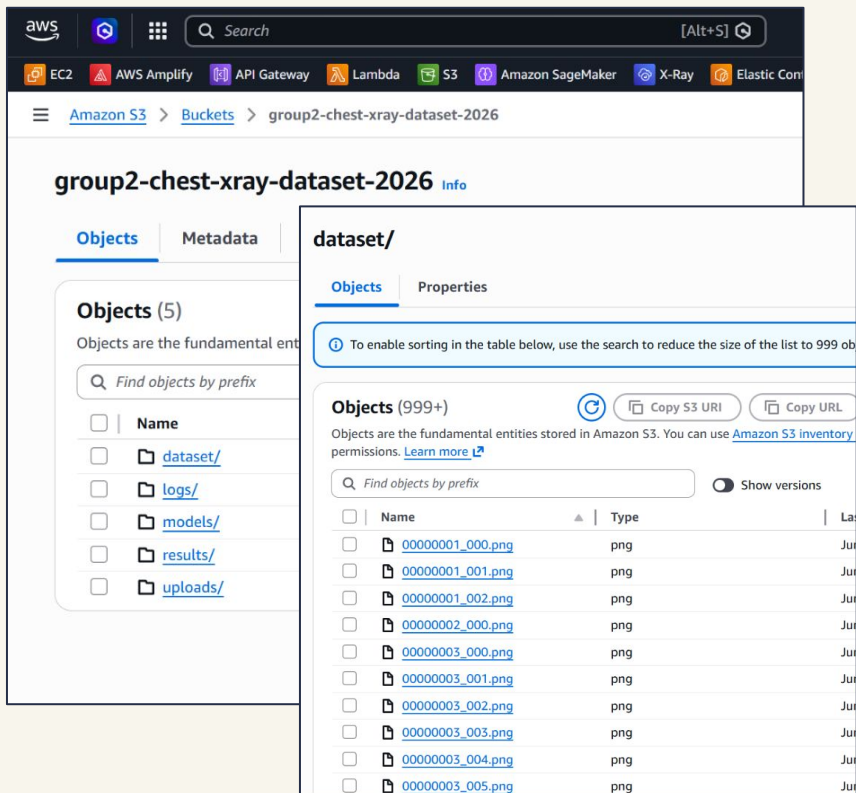
01

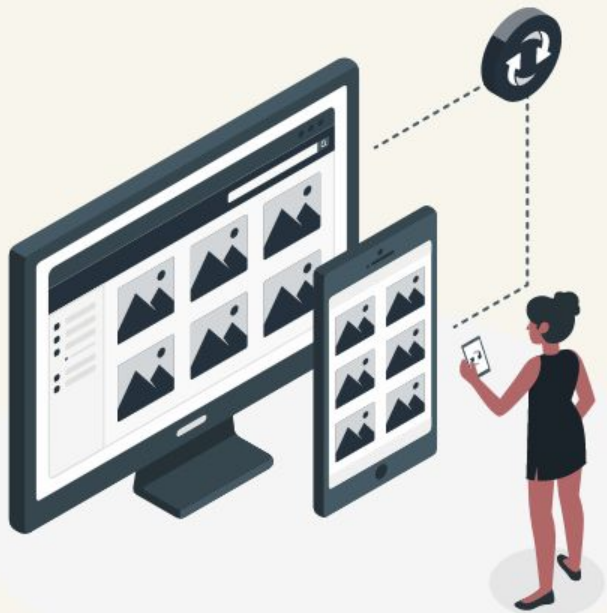
Data & Storage Engineer

Services: S3, EC2, IAM,
CloudWatch, VPC

Storage Platform Integration

- **Transition to Operations:** Shifting focus from initial S3 infrastructure setup to active developer integration
- **Developer Alignment:** Coordinating S3 bucket pathways w/ Alina (Backend) & Kerry (Frontend)
- **Decoupled Architecture:** Establishing clear boundaries to prevent direct, unauthenticated frontend-to-S3 communication



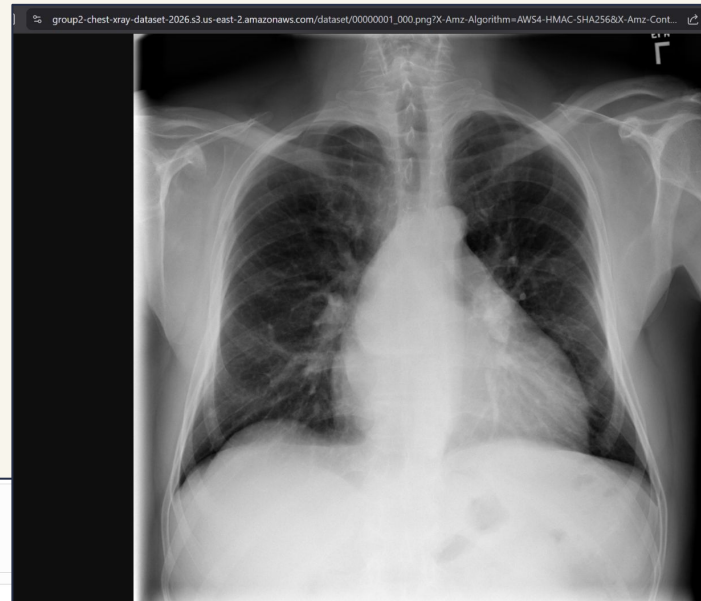


IAM Policy Verification

- **Boundary Testing:** Conducting live tests of the *Group2-Lambda-S3-Execution-Policy* to confirm strict "least privilege" access
- **Directory Isolation:** Verifying backend read/write permissions are limited to */uploads* & */results*
- **Core Protection:** Proving that requests to modify */dataset* or */models* are actively blocked with "Access Denied" errors

Implementing Pre-signed URLs

- **Security First:** Bypassing direct backend compute for file uploads
- **Zero-Credential Exposure:** Eliminates the need for permanent frontend storage credentials
- **Automated TTL:** Enforces a strict 5-minute expiration window on all upload tokens



00000001_000.png [Info](#)

[Properties](#) [Permissions](#) [Versions](#)

Object overview

Owner
5b723f1b731ed5e409ab0b8f627e479e3c54b41870b01314601237c2960ef35d

AWS Region
US East (Ohio) us-east-2

Last modified
June 9, 2026, 17:13:47 (UTC-07:00)

Size
446.5 KB

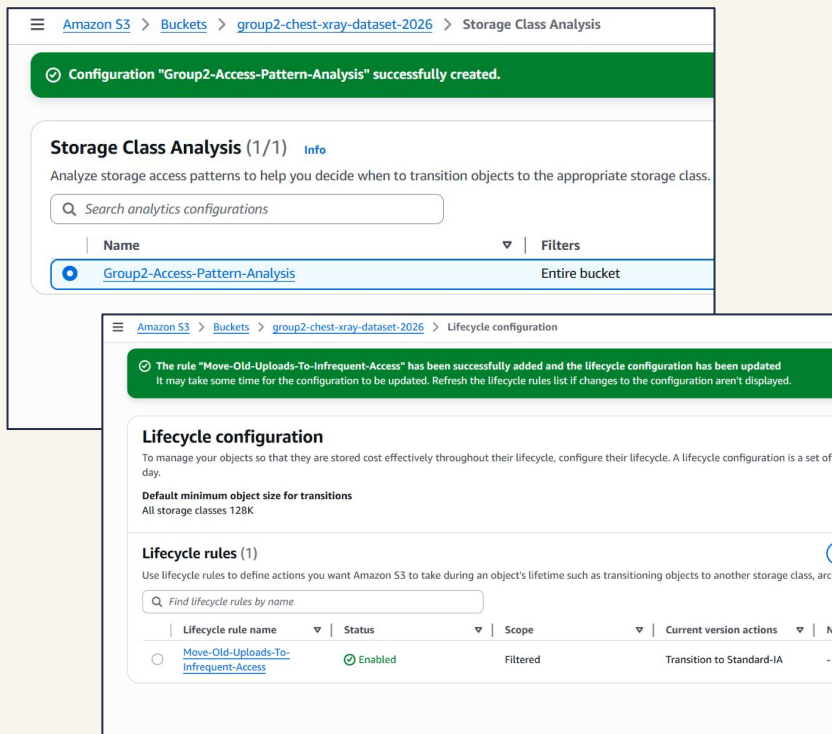
Type
png

S3 URI
s3://group2-chest-xray-dataset-2026/dataset/00000001_000.png

Amazon Resource Name (ARN)
arn:aws:s3::group2-chest-xray-dataset-2026/dataset/00000001_000.png

Entity tag (Etag)
<9ed48cef82776548cf4e074f94c11e5>

Object URL
https://group2-chest-xray-dataset-2026.s3.us-east-2.amazonaws.com/dataset/00000001_000.png



System Observability & Cost Validation

- **Real-Time Metrics:** Validating the 1-minute tracking performance of *Group2-Storage-Activity-Metrics*
- **Access Pattern Analysis:** Monitoring dataset read/write trends using *Group2-Access-Pattern-Analysis*
- **Lifecycle Verification:** Ensuring files in */uploads* automatically transition to cheaper Standard-IA storage after 30 days

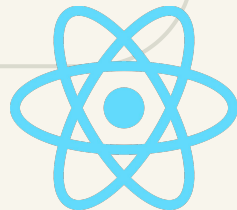


02

Front-End Developer

Service: Amplify

Bridging Backend Services to End Users



- **API Integration:** Implemented API calls to AWS Lambda endpoints for X-ray submission & result retrieval, establishing the critical data pathway between frontend & ML inference pipeline
- **AWS Amplify Deployment:** Developed & deployed fully functional React application through AWS Amplify, creating a production-ready interface for end-user interaction
- **Cross-Team Coordination:** Aligned frontend architecture w/ Alina (Backend API Gateway/Lambda) & Sorea (SageMaker ML model) to ensure seamless data flow and proper request/response handling



Amplify Deployment

- **React App Development:** Built UI using React, leveraging AI tools to optimize code structure
- **Backend API Integration:** Connected the frontend directly to the Lambda REST API for seamless data exchange.
- **GitHub & Environment Tuning:** Managed version control & adjusted code configurations for Amplify compatibility
- **Amplify Hosting:** Deployed live application via Amplify & verified end-to-end functionality

```
EXPLORER
  BONE-FRACTURE-DETECTOR
    build
    static
    {} asset-manifest.json
    {} favicon.ico
    {} index.html
    {} logo192.png
    {} logo512.png
    {} manifest.json
    {} robots.txt
    > node_modules
    > public
    > src
      # App.css
      JS App.js
      JS App.test.js
      # index.css
      JS index.js
      {} logo.svg
      JS postcss.config.js
      JS reportWebVitals.js
      JS setupTests.js
      JS tailwind.config.js
      {} .gitignore
      ! amplify.yml
      {} package-lock.json
      {} package.json
      {} README.md

src > # App.css > @keyframes fadeIn
1
2 {
3   margin: 0;
4   padding: 0;
5   box-sizing: border-box;
6 }
7
8 body {
9   font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
10  background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%);
11  min-height: 100vh;
12  color: #e2e8f0;
13 }
14
15 .app-container {
16   min-height: 100vh;
17   background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%);
18   display: flex;
19   flex-direction: column;
20 }
21
22 /* Header */
23 .app-header {
24   border-bottom: 1px solid #334155;
25   background: rgba(15, 23, 42, 0.5);
26   backdrop-filter: blur(4px);
27   padding: 2rem 1.5rem;
28 }
29
30 .app-header-content {
31   max-width: 48rem;
32   margin: 0 auto;
33 }
34
35 .app-header h1 {
36   font-size: 2.25rem;
37   font-weight: bold;
38   color: white;
39   margin-bottom: 0.5rem;
40 }
```



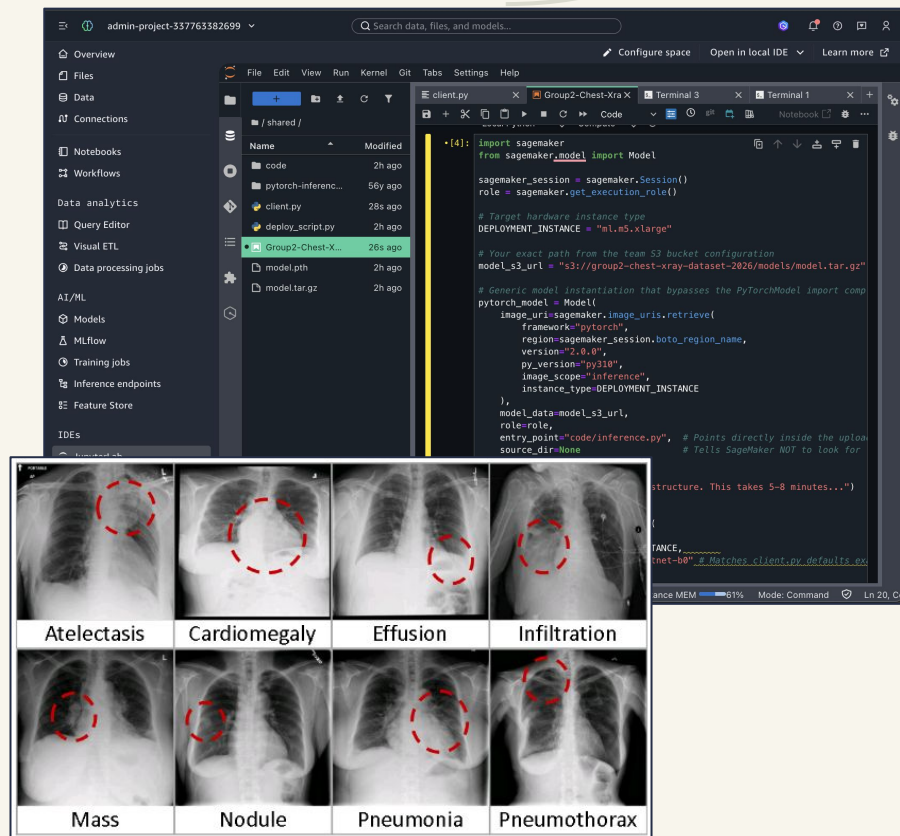
03

ML Engineer

Service: SageMaker

ML Model & Endpoint Deployment

- **Dataset Preparation:** Preprocessed & curated the chest X-ray medical imaging dataset to optimize model training quality
- **EfficientNet-B0 Training:** Fine-tuned a highly efficient convolutional neural network optimized to detect & flag diagnostic anomalies
- **SageMaker Deployment:** Configured & deployed the trained model to a persistent Amazon SageMaker endpoint for real-time inference
- **Diagnostic Predictions:** Programmed pipeline to deliver precise classification outputs accompanied by analytical confidence scores



mura-efficientnet-b0

[View in SageMaker Insights](#)

[Delete](#)

Endpoint summary

Name

mura-efficientnet-b0

ARN

arn:aws:sagemaker:us-east-2:337763382699:endpoint/mura-efficientnet-b0

URL

https://runtime.sagemaker.us-east-2.amazonaws.com/endpoints/mura-efficientnet-b0/invocations

[Learn more about the API](#)

Status

 InService

Creation time

Wed Jun 24 2026 19:43:18 GMT-0700 (Pacific Daylight Time)

Model container logs

[/aws/sagemaker/endpoints/mura-efficientnet-b0](#)

Type

Real-time

Last updated

Wed Jun 24 2026 19:46:12 GMT-0700 (Pacific Daylight Time)

Alarms

0 alarms



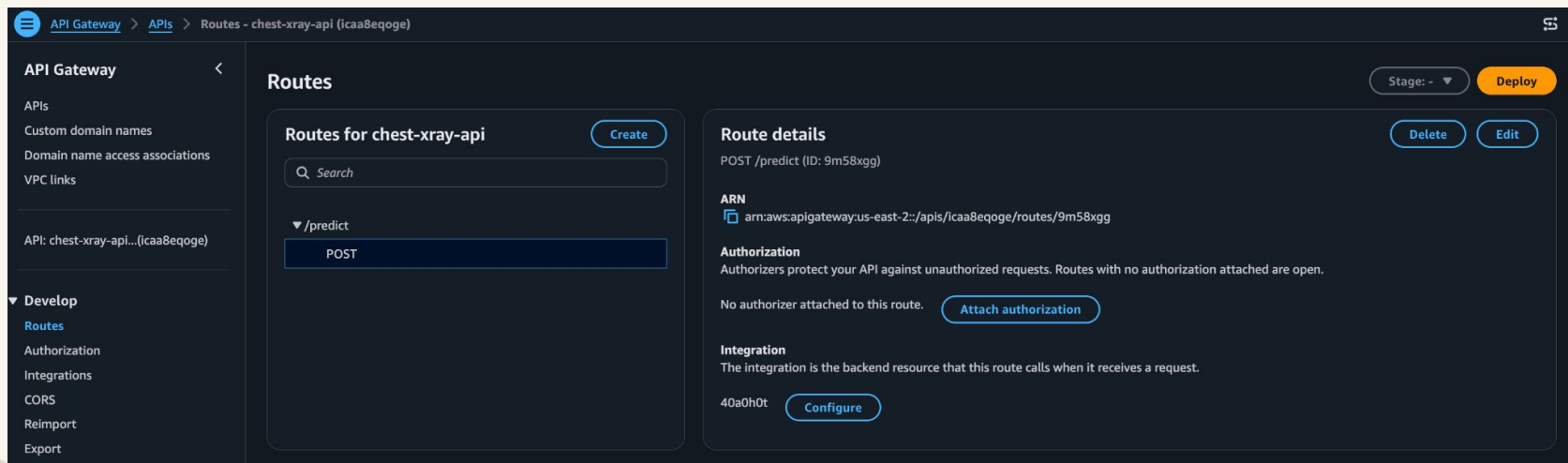
04

Backend Engineer

Services: Lambda, API
Gateway, IAM

Cloud Infrastructure & Backend

- **API & Serverless Integration:** Built API Gateway REST API & integrated it w/ AWS Lambda to bridge frontend & backend
- **Secure Image Pipeline:** Enabled frontend to safely transmit uploaded X-ray images to Lambda for automated processing.



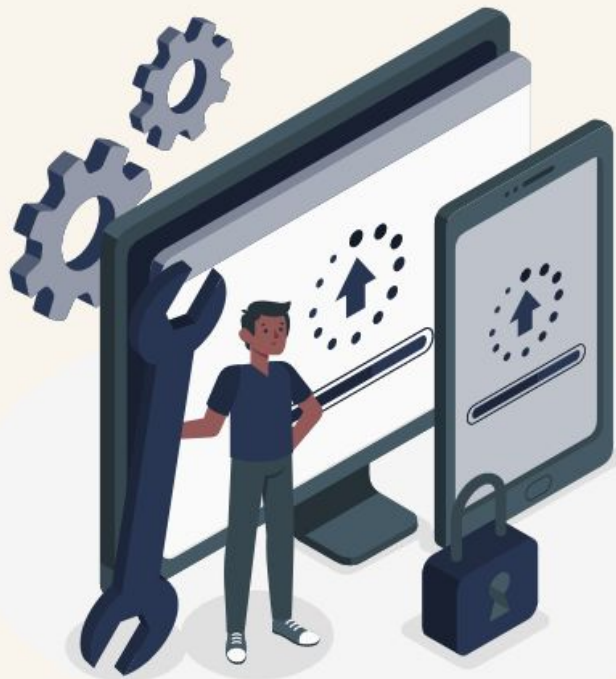
Cloud Infrastructure & Backend

- **SageMaker Orchestration:** Configured Lambda to handle incoming data requests & communicate directly w/ machine learning model
- **IAM Security Framework:** Established granular IAM roles & permissions to ensure secure communication across all AWS services

The screenshot displays the AWS Lambda console interface. The main panel shows the 'Code source' for the 'chest-xray-predict-function'. The code is a Python lambda function that handles CORS preflight requests. The function returns a 200 status code and headers for 'Access-Control-Allow-Origin', 'Access-Control-Allow-Headers', and 'Access-Control-Allow-Methods'. The 'TEST EVENTS' section shows a successful test event with a response body indicating 'No Concern Detected' and a confidence of 100.

On the right, the 'Cross-Origin Resource Sharing' configuration is shown. It includes the following settings:

- Access-Control-Allow-Origin:** `https://main.d2rssidn4duqfs.amplifyapp.com/`
- Access-Control-Allow-Headers:** `content-type,x-amz-date,authorization,x-api-key,x-amz-security-token`
- Access-Control-Allow-Methods:** `POST`, `OPTIONS`
- Access-Control-Expose-Headers:** `https://main.d2rssidn4duqfs.amplifyapp.com/`
- Access-Control-Max-Age:** 0 seconds
- Access-Control-Allow-Credentials:** NO



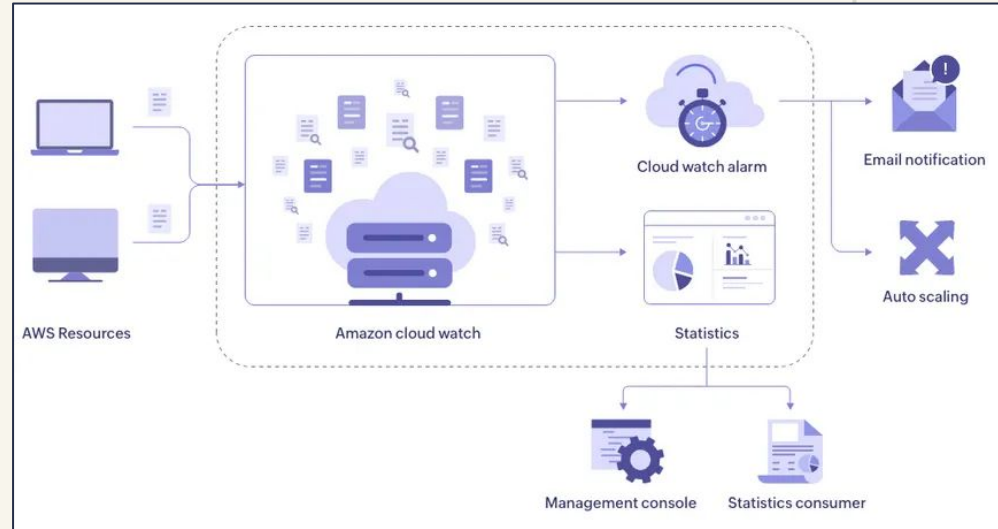
05

QA Tester & Orchestrator

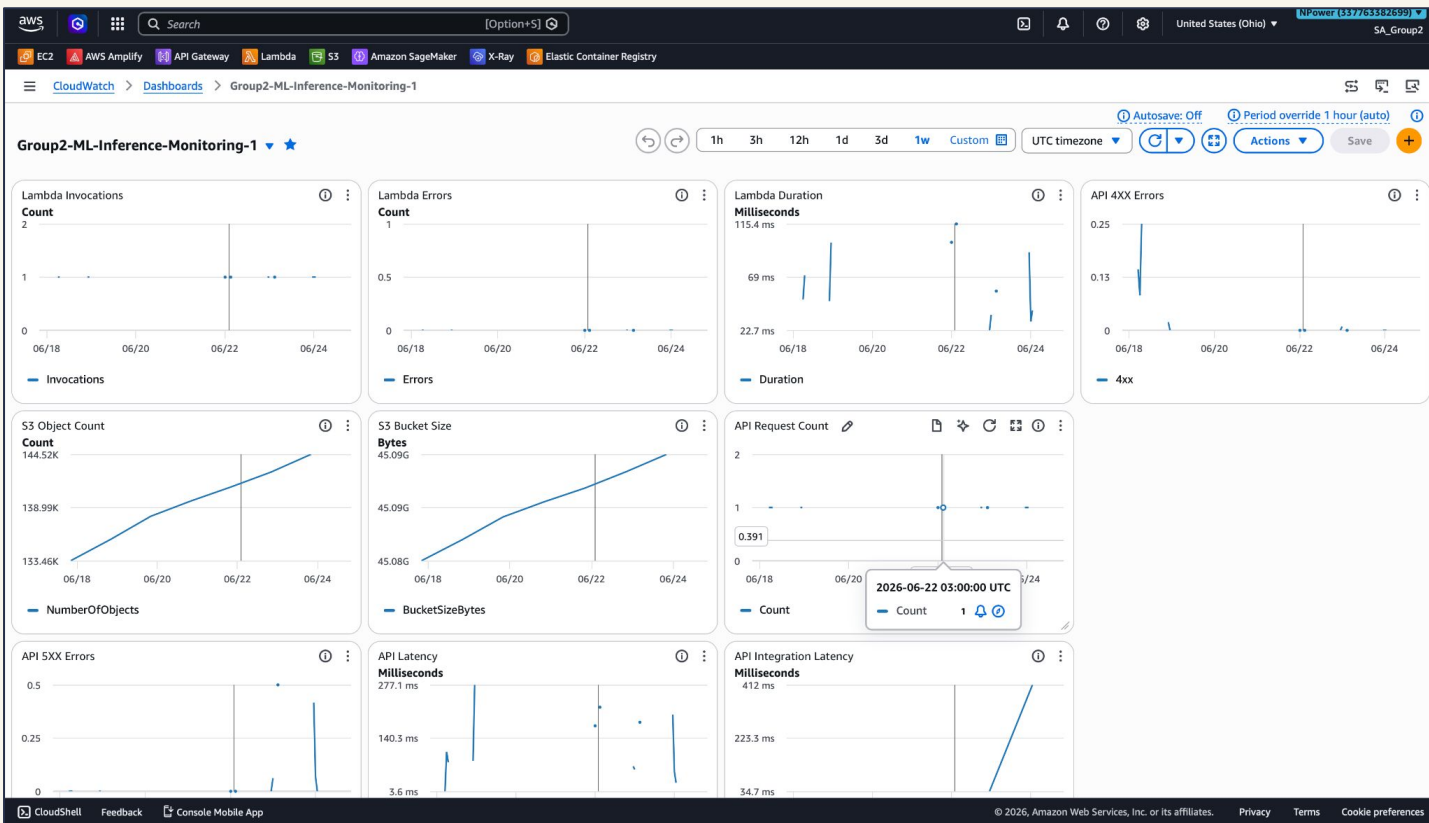
Service: CloudWatch

Why CloudWatch Matters?

- **Eliminating Blind Spots:** Real-time tracking of API, Lambda, & ML model performance
- **Centralized Health Pulse:** Unified dashboarding for proactive issue identification
- **Root-Cause Acceleration:** Drastically reducing diagnostic time for system bottlenecks



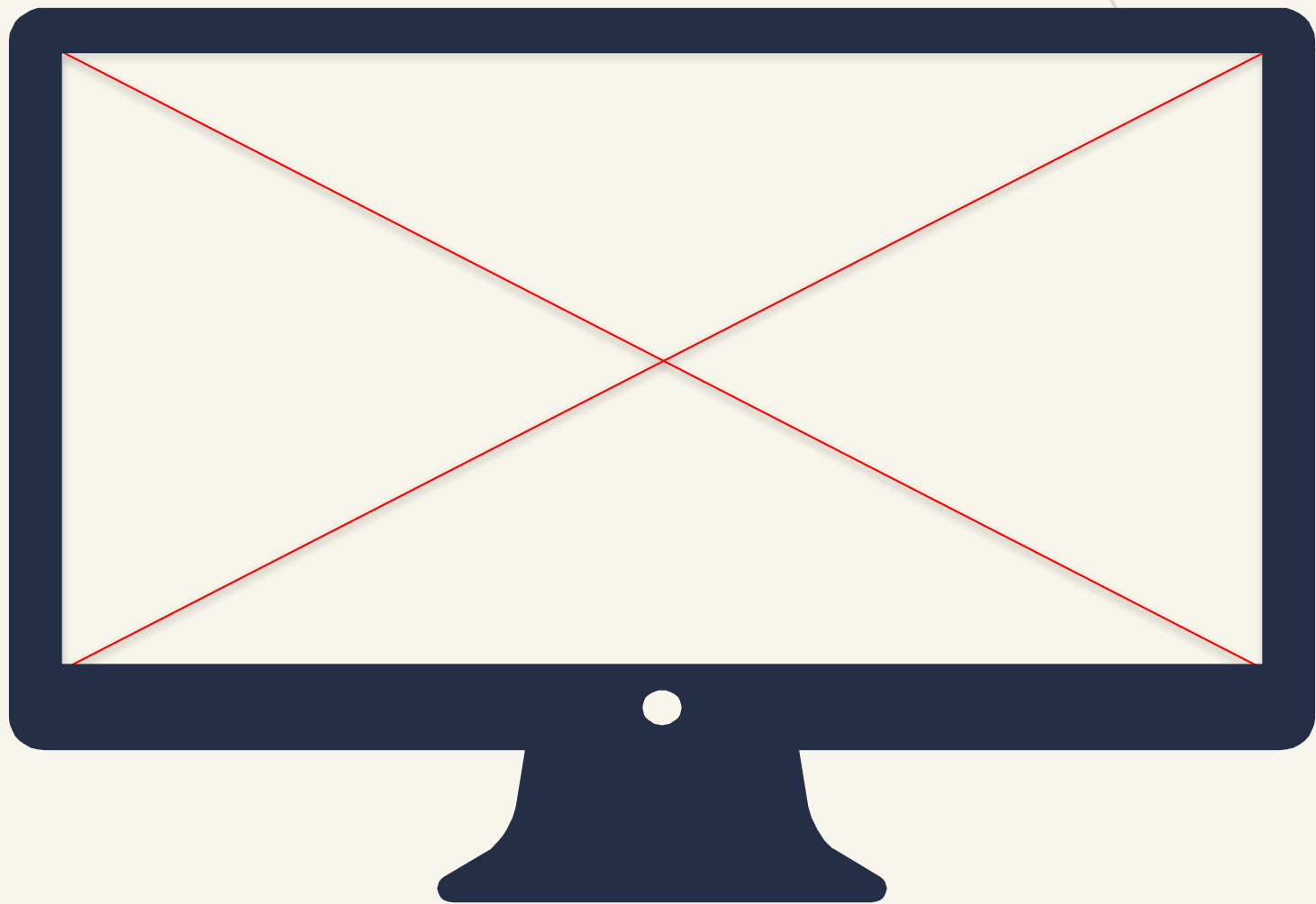
Live Dashboard



03

Demo





Results

Bone Fracture Detector

AI-powered X-ray analysis

Analysis Results

CLASSIFICATION

● Possible Concern Detected

FINDING TYPE

test result only

CONFIDENCE SCORE

85.0%

This score indicates how confident the AI model is in its prediction. A higher percentage means the model is more certain about the result.

Score Interpretation:

- 90-100%: Very confident in the prediction
- 75-89%: Moderately confident
- 50-74%: Less certain (borderline case)

Important: Even with high confidence, this is a screening tool only. Always consult a licensed medical professional for diagnosis.

MESSAGE

⚠ This is not a diagnosis. Please follow up with a licensed medical professional.

ANALYZED IMAGE



⏮ Analyze Another

Bone Fracture Detector

AI-powered X-ray analysis

Analysis Results

CLASSIFICATION

● No Concern Detected

FINDING TYPE

CONFIDENCE SCORE

NaN%

This score indicates how confident the AI model is in its prediction. A higher percentage means the model is more certain about the result.

Score Interpretation:

- 90-100%: Very confident in the prediction
- 75-89%: Moderately confident
- 50-74%: Less certain (borderline case)

Important: Even with high confidence, this is a screening tool only. Always consult a licensed medical professional for diagnosis.

MESSAGE



ANALYZED IMAGE



⏮ Analyze Another

Thank You

Sincere thanks to **Ernesto** and **Andrew** for the incredible support that made our AWS Cloud Practitioner and Solutions Architect milestones a reality.

Any questions?

github.com/sorea-asmai/Chest-Detect